



JOBSHEET I

BASIC PROGRAMMING CONCEPTS

1. Purpose of the Practicum

After doing this practicum material, students will be able to:

1. Implement selection, looping, arrays, and functions in Java program code

2. Practicum

2.1 Selection

Trial time: 50 minutes

The material in this practicum has been explained in the Basic Programming course. So that in this practicum, no experimental steps will be carried out. Answer the following questions:

2.2.1 Election Practicum

Questions

1. Create a program to calculate the final score of students with the provision of 20% of assignment scores, 20% of quiz scores, 30% of UTS scores, and 30% of UAS scores. Each value entered has a value limit of 0 - 100. When a user enters outside of that range, an "invalid value" will come out. When the final score has been obtained, then convert the score with the following conditions:

Nilai Angka	Nilai Mutu		
	Nilai Huruf	Nilai Setara	Kualifikasi
$80 < N \leq 100$	A	4	Sangat Baik
$73 < N \leq 80$	B+	3,5	Lebih dari Baik
$65 < N \leq 73$	B	3	Baik
$60 < N \leq 65$	C+	2,5	Lebih dari Cukup
$50 < N \leq 60$	C	2	Cukup
$39 < N \leq 50$	D	1	Kurang
$N \leq 39$	E	0	Gagal

If the Letter Values obtained are A, B+, B, C+, C then PASS, if the letter values D and E then NOT PASS.

- Inputs from the program in the form of assignment value components, quizzes, UTS, UAS
- Output from the "invalid value" program if the value entered is outside the provisions
- The output of the program is in the form of final grades, letter grades, and the description of PASS/NOT PASS



Example of program running results

```
Program Menghitung Nilai Akhir
=====
Masukkan Nilai Tugas: 85
Masukkan Nilai Kuis: 90
Masukkan Nilai UTS: 120
Masukkan Nilai UAS: 70
=====
nilai tidak valid
=====
```

```
Program Menghitung Nilai Akhir
=====
Masukkan Nilai Tugas: 90
Masukkan Nilai Kuis: 40
Masukkan Nilai UTS: 75
Masukkan Nilai UAS: 85
=====
nilai akhir : 74.0
Nilai Huruf :B+
=====
SELAMAT ANDA LULUS
```

2.2 Looping

Trial time: 50 minutes

The material in this practicum has been explained in the Basic Programming course. So that in this practicum, no experimental steps will be carried out. Answer the following questions:

2.3.1 Repetition Practicum

Questions

1. Create a program that can display a row of numbers from 1 to n, with **n = the last 2-digit** of your NIM. Here are the conditions for printing the series
 - a. Number of multiples of 3 printed with symbols #
 - b. Even numbers other than multiples of 3 are printed according to the number
 - c. Odd numbers are printed with the symbol *
 - d. Numbers 10 and 15 are not printed

*if $n < 10$ then add 10 ($n += 10$)

Example 1: NIM input: 2541720102 then $n=12$

Output: *2#4*#8*#

Example 2: NIM input: 2541720120 then $n=20$

Output: *2#4*#8*#*14*16*#20

2.3 Array

Trial time: 50 minutes

The material in this practicum has been explained in the Basic Programming course. So that in this practicum, no experimental steps will be carried out. Answer the following questions:

2.4.1 Array Practicum

Questions

1. Create a program to calculate the Semester IP of the courses you took last semester. The formula for calculating the semester IP is as follows:

$$IP \text{ Semester} = \frac{\sum_i (\text{Nilai Setara}_i * \text{bobot SKS}_i)}{\sum SKS}$$

Equivalent values are obtained from the following conversion table:

Nilai Angka	Nilai Mutu		
	Nilai Huruf	Nilai Setara	Kualifikasi
80 < N ≤ 100	A	4	Sangat Baik
73 < N ≤ 80	B+	3,5	Lebih dari Baik
65 < N ≤ 73	B	3	Baik
60 < N ≤ 65	C+	2,5	Lebih dari Cukup
50 < N ≤ 60	C	2	Cukup
39 < N ≤ 50	D	1	Kurang
N ≤ 39	E	0	Gagal

Input from the program is in the form of **course name**, **credit weight**, and **letter grades** of the course.

Example of program running results:

```

=====
Program Menghitung IP Semester
=====
masukkan nilai Angka untuk MK Pancasila: 75
masukkan nilai Angka untuk MK Konsep Teknologi Informasi: 85
masukkan nilai Angka untuk MK Critical Thinking dan Problem Solving: 70
masukkan nilai Angka untuk MK Matematika Dasar: 85
masukkan nilai Angka untuk MK Bahasa Inggris: 85
masukkan nilai Angka untuk MK Dasar Pemrograman: 62
masukkan nilai Angka untuk MK Praktikum Dasar Pemrograman: 62
masukkan nilai Angka untuk MK Keselamatan dan Kesehatan Kerja: 85
=====
hasil Konversi Nilai
=====
MK                               Nilai Angka   Nilai Huruf   Bobot Nilai
Pancasila                        75.00         B+            3.50
Konsep Teknologi Informasi       85.00         A             4.00
Critical Thinking dan Problem Solving 70.00         B             3.00
Matematika Dasar                 85.00         A             4.00
Bahasa Inggris                   85.00         A             4.00
Dasar Pemrograman                62.00         C+            2.50
Praktikum Dasar Pemrograman      62.00         C+            2.50
Keselamatan dan Kesehatan Kerja   85.00         A             4.00
=====
IP : 3.42

```

2.4 Functions

Trial time: 50 minutes



The material in this practicum has been explained in the Basic Programming course, so in this practicum, no experimental steps will be carried out. Answer the following questions:

2.5.1 Functional Practicum

Questions

RoyalGarden is a florist that has many branches. Every day the Flower Stock and the flowers sold are always recorded with the following details:

Row = Store Branch, Column = Flower stock on day x

	Aglonema	Yam	Alocasia	Rose
RoyalGarden 1	10	5	15	7
RoyalGarden 2	6	11	9	12
RoyalGarden 3	2	10	10	5
RoyalGarden 4	5	7	12	9

Details of the price of Aglonema = 75,000, Yam = 50,000, Alocasia =60,000, Roses =10,000.

1. Create a function to display the income of each branch if all the flowers are sold out.
2. Show the status of each branch with the following conditions:
 - a. If the branch's income > IDR 1,500,000, then the branch gets the status of "Very Good"
 - b. If the income $\leq$ IDR 1,500,000, the status is "Need Evaluation".

3. Duties

Processing time: 100 minutes

1. Arrange the program to create two arrays and their contents as follows. The first array is a one-dimensional array of char CODE[10], containing the car's license plate code. The second array, the two-dimensional array char KOTA[10][12] contains the name of the city paired with the car's license plate code. The illustration of the display of the array is as follows:



A	B	A	N	T	E	N						
B	J	A	K	A	R	T	A					
D	B	A	N	D	U	N	G					
E	C	I	R	E	B	O	N					
F	B	O	G	O	R							
G	P	E	K	A	L	O	N	G	A	N		
H	S	E	M	A	R	A	N	G				
L	S	U	R	A	B	A	Y	A				
N	M	A	L	A	N	G						
T	T	E	G	A	L							

When the user provides a license plate code input, the program will remove the city name from the license plate code.

2. A program is used to store and manage a student's lecture schedule. Schedule data is stored in a string-type 2-dimensional array, provided that

- The line states the i-i schedule
- The column states schedule information: Course Name, Room, Lecture Day, Lecture Hours. Example :

```
schedule[0][0] = "Basic Programming"
```

```
schedule[0][1] = "Computing Lab 1"
```

```
schedule[0][2] = "Monday"
```

```
schedule[0][3] = "8:00–10:00"
```

- The number of lecture schedules is **n**, input by the user.

Create a function for

- a. Input lecture schedule data into a 2-dimensional array
- b. Display the entire lecture schedule in the form of a table
- c. Display lecture schedules based on specific days
- d. Display the lecture schedule by the name of a specific course